

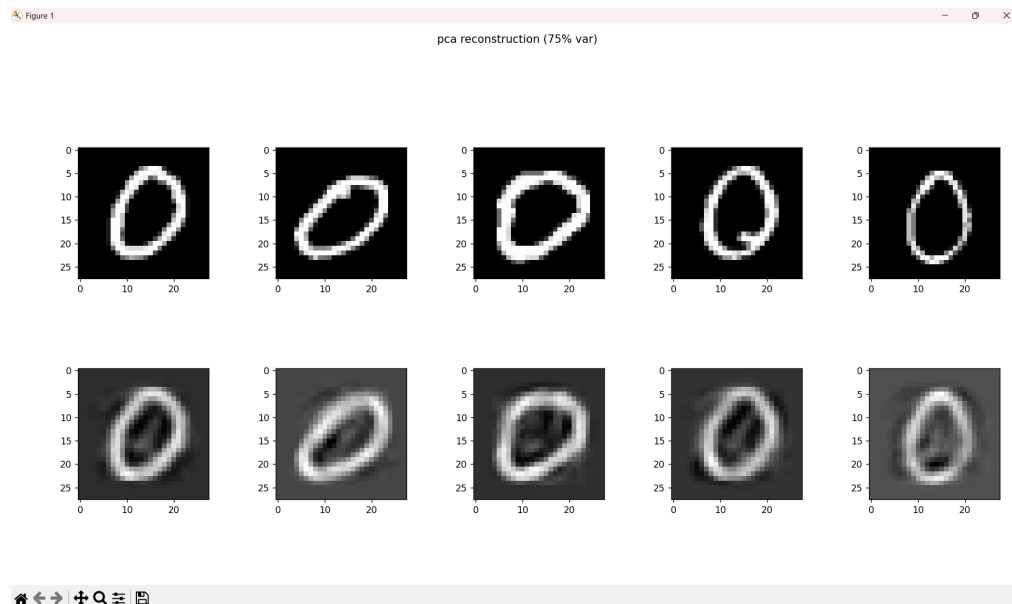
SML A2 Visualisation

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PCA Reconstruction

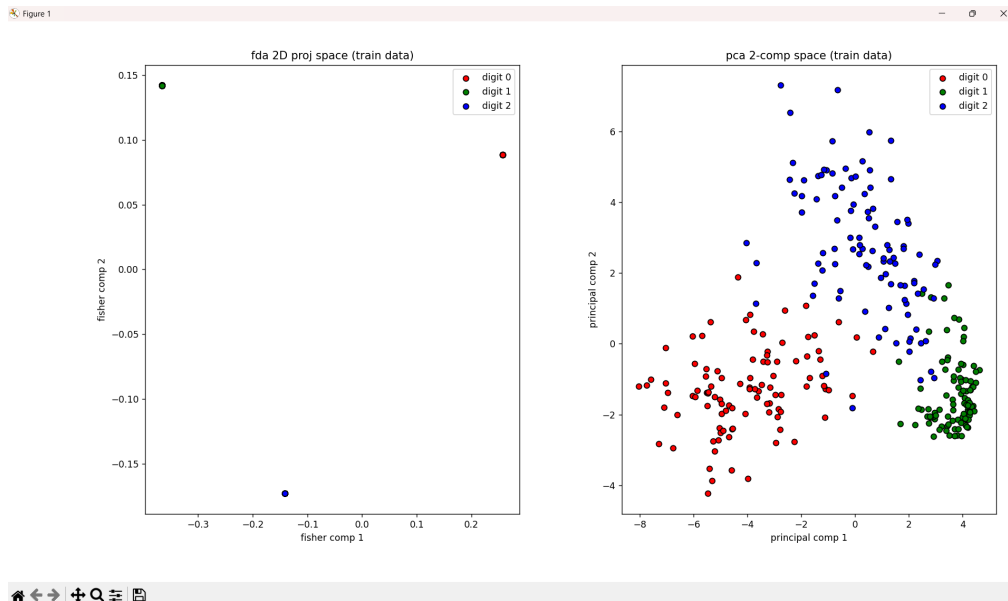
I kept enough PCA features to keep 75% of the variance, and then rebuilt the first 5 training images back into their original 784-pixel form to check how good the compression was.



Top row: original images.
Bottom row: reconstructed images.
The reconstructions keep the main digit shape but look slightly blurry because some components were removed.

Feature Space Visualisation

The scatter plots below show the training data projected into 2D using FDA (left) and PCA using only the first two principal components (right).



Left: FDA 2D projection space.
 Right: PCA 2-component projection space.
 Each point represents one training sample, coloured by class.

Observations

1. **FDA projection:** The three digit classes appear in clearly separate areas in the Fisher space. This happens because FDA tries to spread different classes apart while keeping points from the same class close together, making them easy to tell apart even in 2D.
2. **PCA projection:** Digit 1 (green) forms a small, clear cluster. However, digits 0 and 2 overlap more. This is because PCA does not use class labels when choosing directions, so it doesn't try to separate the classes.